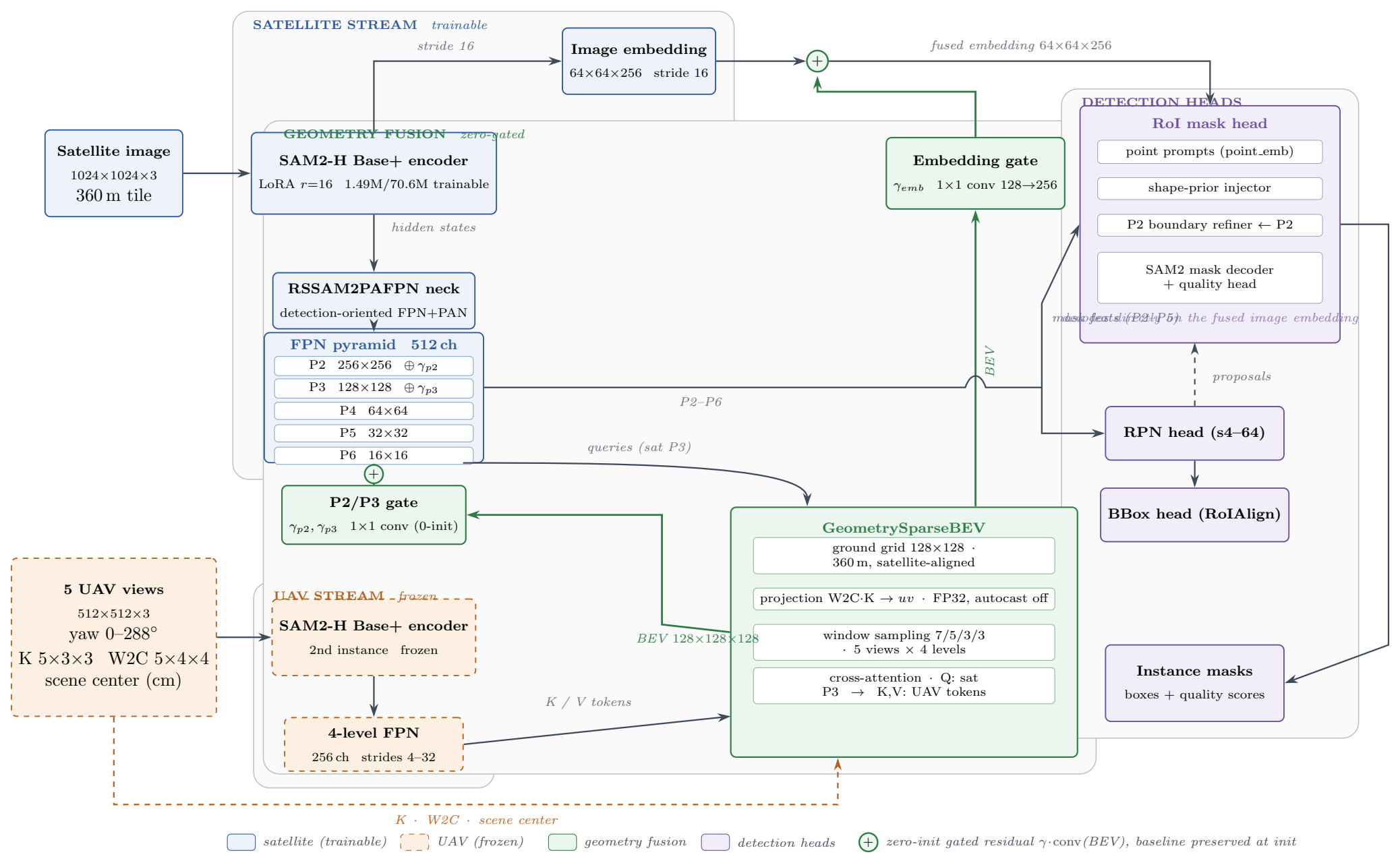

SATELLITE STREAM trainable
stride 16
Image embedding
64×64×256 stride 16
fused embedding 64×64×256
Satellite image
1024×1024×3
360 m tile
GEOMETRY FUSION zero-gated
SAM2-H Base+ encoder
LoRA r=16 1.49M/70.6M trainable
hidden states
RSSAM2PAFPN neck
detection-oriented FPN+PAN
FPN pyramid 512 ch
P2 256×256 ⊕ γp2
P3 128×128 ⊕ γp3
P4 64×64
P5 32×32
P6 16×16
P2–P6
queries (sat P3)
P2/P3 gate
γp2, γp3 1×1 conv (0-init)
Embedding gate
γemb 1×1 conv 128→256
BEV
DETECTION HEADS
RoI mask head
point prompts (point_emb)
shape-prior injector
P2 boundary refiner ← P2
SAM2 mask decoder
+ quality head
mask feats (P2–P5)
proposals
RPN head (s4–64)
BBox head (RoIAlign)
GeometrySparseBEV
ground grid 128×128 ·
360 m, satellite-aligned
projection W2C·K → uv · FP32, autocast off
window sampling 7/5/3/3
· 5 views × 4 levels
cross-attention · Q: sat
P3 → K,V: UAV tokens
BEV 128×128×128
Instance masks
boxes + quality scores
5 UAV views
512×512×3
yaw 0–288°
K 5×3×3 W2C 5×4×4
scene center (cm)
UAV STREAM frozen
SAM2-H Base+ encoder
2nd instance frozen
4-level FPN
256 ch strides 4–32
K / V tokens
K · W2C · scene center
satellite (trainable)
UAV (frozen)
geometry fusion
detection heads
zero-init gated residual γ·conv(BEV), baseline preserved at init